Customer Shopping Behavior Analysis

Uncovering strategic insights from 3,900 customer purchases to guide business decisions.

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PROJECT SCOPE

Analyzing 3,900 Transactions

This project analyzes customer shopping behavior using transactional data across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior.

Spending Patterns

Customer Segments

Product Preferences

Subscription Behavior

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Dataset Summary: Key Features

The dataset contains 3,900 rows and 18 columns, capturing a wide range of customer and purchase details.

Customer Demographics

Age, Gender, Location, Subscription Status.

Purchase Details

Item Purchased, Category, Purchase Amount, Season, Size, Color.

Shopping Behavior

Discount Applied, Promo Code Used, Previous Purchases, Review Rating, Shipping Type.

Note: 37 missing values were identified in the Review Rating column and imputed using the median rating per product category.

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Exploratory Data Analysis: Python Workflow

Data preparation and cleaning involved standardization, imputation, and feature engineering.

Data Loading & Exploration

Imported dataset (pandas), checked structure (df.info()), and summary statistics (.describe()).

Data Cleaning

Handled 37 missing values in Review Rating using median imputation by category.

Standardization & Engineering

Renamed columns to snake case. Created `age_group` *and* `purchase_frequency_days`.

Consistency Check

Verified and dropped redundant `promo_code_used` *column.*

Database Integration

Loaded the cleaned DataFrame into PostgreSQL for structured SQL analysis.

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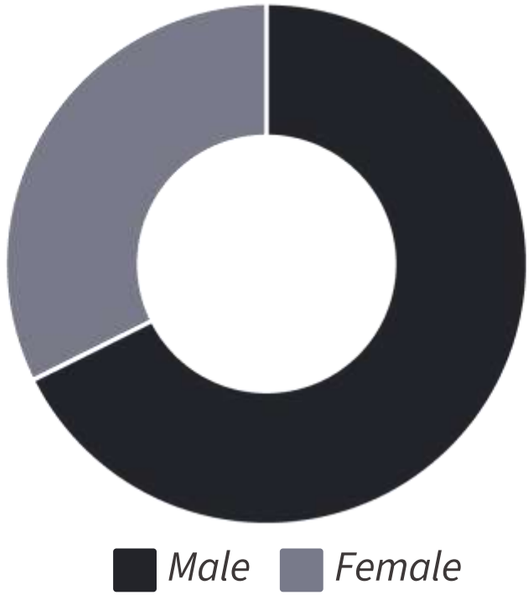
SQL Analysis: Revenue & Gender

Male customers generate significantly higher total revenue compared to female customers.



$157.9K

Male Revenue

Total revenue generated by male customers.

$75.2K

Female Revenue

Total revenue generated by female customers.

This disparity suggests a need for targeted marketing strategies to increase female customer spend.

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Key Product Insights

Analysis of product performance based on ratings, discounts, and category sales.

Top 5 Products by Average Rating

- ***Gloves:*** *3.86*
- ***Sandals:*** *3.84*
- ***Boots:*** *3.82*
- ***Hat:*** *3.80*
- ***Skirt:*** *3.78*

Discount-Dependent Products

Products with the highest percentage of discounted purchases:

- ***Hat:*** *50.00%*
- ***Sneakers:*** *49.66%*
- ***Coat:*** *49.07%*

Customer Segmentation & Loyalty

Customers were classified based on purchase history, revealing a high proportion of loyal buyers.

Loyal	Returning	New
3,116 Customers	*701 Customers*	*83 Customers*

Repeat buyers (customers with >5 purchases) are significantly more likely to subscribe:

Subscription Status	*Repeat Buyers*
No	*2,518*
Yes	*958*

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Subscription & Shipping Comparison

Subscribers have a slightly lower average spend than non-subscribers, but Express Shipping users spend more.

Subscribers vs. Non-Subscribers

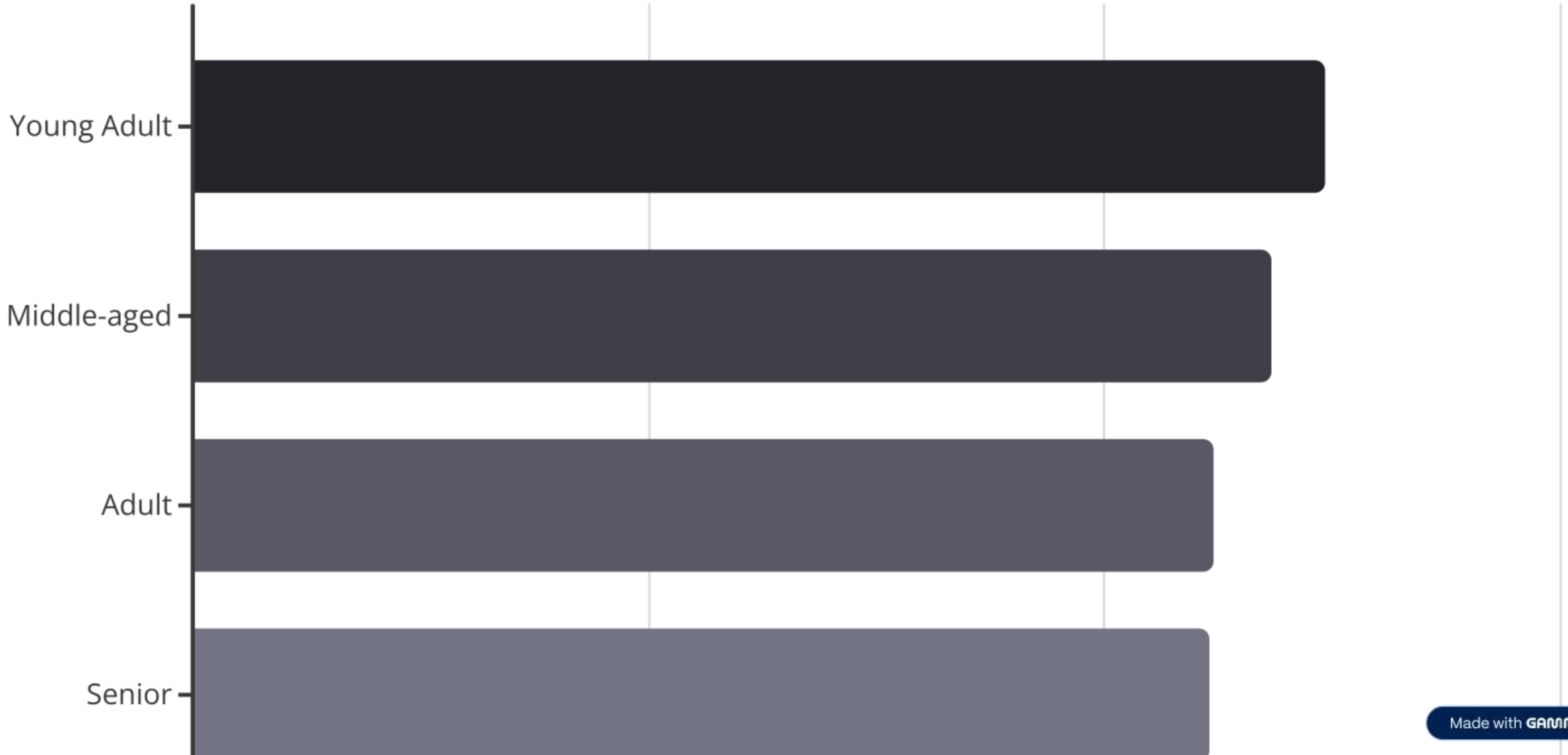
Status	Avg. Spend
Yes (1,053 customers)	$59.49
No (2,847 customers)	$59.87

Shipping Type Spend

Shipping Type	Avg. Purchase
Standard	$58.46
Express	$60.48

Revenue Contribution by Age Group

Young Adults and Middle-aged customers are the highest revenue contributors.



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Strategic Business Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber value and retention.

Customer Loyalty Programs

Reward repeat buyers to transition them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control, especially for discount-dependent products.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users for higher conversion.

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